

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Define distillation processes, focusing on atmospheric Distillation Unit (ADU) and Vacuum Distillation Unit (VDU). And also explain with a diagram the components and flow of both units.
- Q.24 With neat and clean flow sheet, explain the manufacturing of ammonia, including the chemical reactions, key raw materials, and process description.
- Q.25 Provide a detailed process description of Portland cement production, incorporating with flow sheet.

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Roll No.

3rd Sem / Chemical (Pulp & Paper)

Subject : Process Industries

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the main raw material for producing urea?
- a) Nitrogen b) Phosphoric acid
c) Ammonia d) Potassium chloride
- Q.2 What is the main purpose of the bleaching process in sugar manufacturing?
- a) Increase sweetness
b) Remove color
c) Increase yield
d) Improve crystallization
- Q.3 Atmospheric Distillation Unit (ADU) is used to:
- a) Remove sulfur
b) Separate crude oil into fractions
c) Dehydrate crude oil
d) Increase octane number

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Q.4 Which property is measured using a viscometer?

- a) Flash point b) Viscosity
- c) Specific Gravity d) Smoke Point

Q.5 Cement is classified as:

- a) A soil b) A binder
- c) A filler d) None of above

Q.6 The flash point of a fuel is important because it indicates:

- a) Its boiling point
- b) Its combustion quality
- c) The temperature at which it ignites
- d) Its viscosity

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is crystallization.

Q.8 Write the formula for specific gravity.

Q.9 Which component is removed during the desulfurization process in petroleum refining?

Q.10 What is the purpose of the filtration process in sugar technology

Q.11 Write the one use of Fertilizer

Q.12 Draw the diagram of Filtration.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the significance of NPK fertilizers.

Q.14 Differentiate between settling process and sedimentation?

Q.15 Write down about reforming and its types.

Q.16 Write down about Desulphurization and dehydration?

Q.17 Explain the manufacturing process of super phosphate(SSP)

Q.18 Write a note on origin of petroleum.

Q.19 Write down about significance of smoke point and aniline point

Q.20 Describe the process of urea production.

Q.21 Discuss about the difference between cloud point and pour point.

Q.22 Describe about the manufacturing of sugar with flow sheet and process description .